

Business Analytics

2024-10-01 to 2026-09-30 · Demo Client

Prepared for	Demo Client
Engagement	DEMO-0011
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Executive summary

Demonstration report. Built entirely from synthetic data generated by ProQuant for illustration: a café-restaurant chain that does not exist. No real business, customer or person is described.

ProQuant Analytics brought together every record Demo Client keeps (expenses, items, marketing, customers, places, staff, targets, transactions) for 2024-10-01 to 2026-09-30, and asked where the business makes and loses money, what moves its sales, and what to do next. The data came in the client's own formats and was mapped to one standard set of tables; what was left out, and why, is listed in section 11.

Overall view: The business needs attention

7 of 17 health tests passed, 6 with a warning, 4 failed. The thresholds were agreed before the results were seen.

SALES, LAST 12 MONTHS
SAR 4.98m
+23.7% recently, against the same months a year earlier

GROSS MARGIN
60.6%
sales less the cost of what was sold

OPERATING PROFIT
SAR -395k
-7.9% of sales, after head office

ORDERS, 12 MONTHS
67,022
3 branches

NEW GUESTS WHO RETURN
89%
within 90 days

NEXT SIX MONTHS
SAR 3.19m
range 2.40m to 4.24m

Findings

- **Sales** Sales were SAR 4.98m over the last 12 months; the recent months are +23.7% against the same months a year earlier.
- **Profit** After every cost, including head office, the business lost SAR 395k (-7.9% of sales).
- **Branches** Loss-making branches: Airport, with 36% of sales.
- **Dishes** Popular but low margin: Family Platter.
- **Guests** 89% of new guests come back within 90 days; a new one brings SAR 1,115 in a year.
- **Prices** After the +16% price change on Mains in 2025-10, units moved -7%.
- **Marketing** Snapchat took 43% of the spend and returned 0.68 of gross profit per unit spent.
- **Outlook** Next six months: about SAR 3.19m of sales (range SAR 2.40m to SAR 4.24m).
- **Analyst** Illustrative note (synthetic data): the Airport lease runs to 2027 with a break clause in March; the owner asked for the cost of leaving early.

Actions

Ref	Priority	Action	Owner	Due
A-01	High	Win back the customers lost since the price rise; review the rise on the items that moved most.	Marketing	2026-11-04
A-02	High	Reprice or redesign the popular low-margin items.	Operations manager	2026-11-04
A-03	High	Agree a profit plan: start with the loss-making sites and the low-margin items.	Owner	2026-11-04
A-04	High	Decide on each loss-making site: renegotiate the rent, cut costs, or close.	Owner	2026-11-04
A-05	Medium	Recover the missing days from the system or confirm the site was closed.	Data owner	2026-12-04
A-06	Medium	Add the missing items and their costs to the item list.	Finance	2026-12-04
A-07	Medium	Move spend from the weakest channel to the best one; review in a month.	Marketing	2026-12-04
A-08	Medium	Match staff hours to the busy hours at the weakest site.	Operations manager	2026-12-04
A-09	Medium	Reset the targets or agree a plan with the site managers who miss them.	Finance	2026-12-04
A-10	Medium	Use the forecast range, not the single number, until the error falls.	Finance	2026-12-04

Source: tables/actions.csv

Methodology

This report tells what was found and what it means for the business. How each number is built is in the methodology document; the definition of every measure is in the measure catalogue. One thing helps when reading on: profit by branch counts every cost booked to that branch, while head-office costs are shown on their own and never spread over the branches.

Results

1. Sales and the months ahead

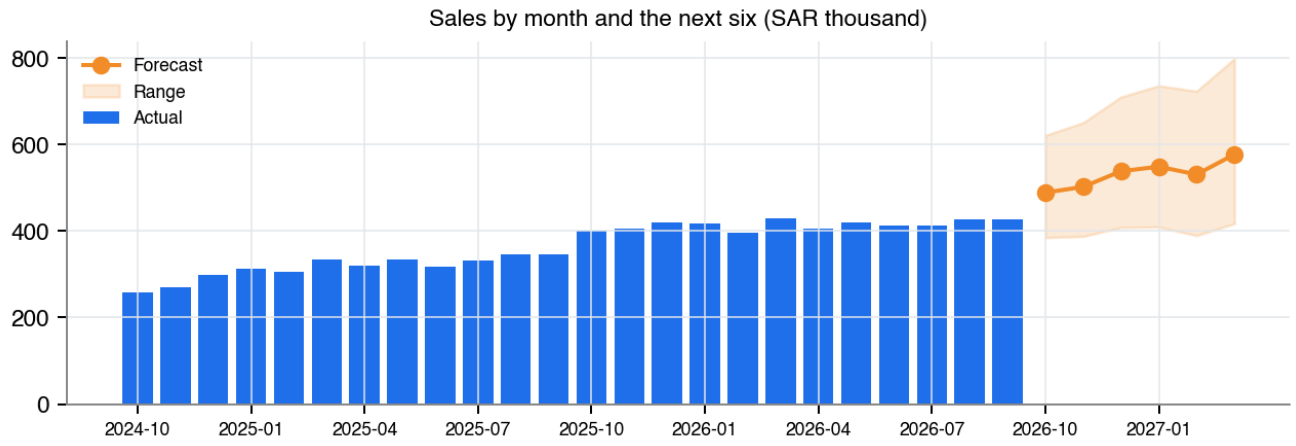


Figure 1. Sales by month, with the forecast and its range (trend and season). Source: tables/forecast.csv

Sales over the last twelve months were SAR 4.98m. The latest three months are +23.7% against the same months a year earlier. Held back and forecast, the last 6 months were missed by 10.9% on average, which is why the range matters more than the single line.

2. What moved sales

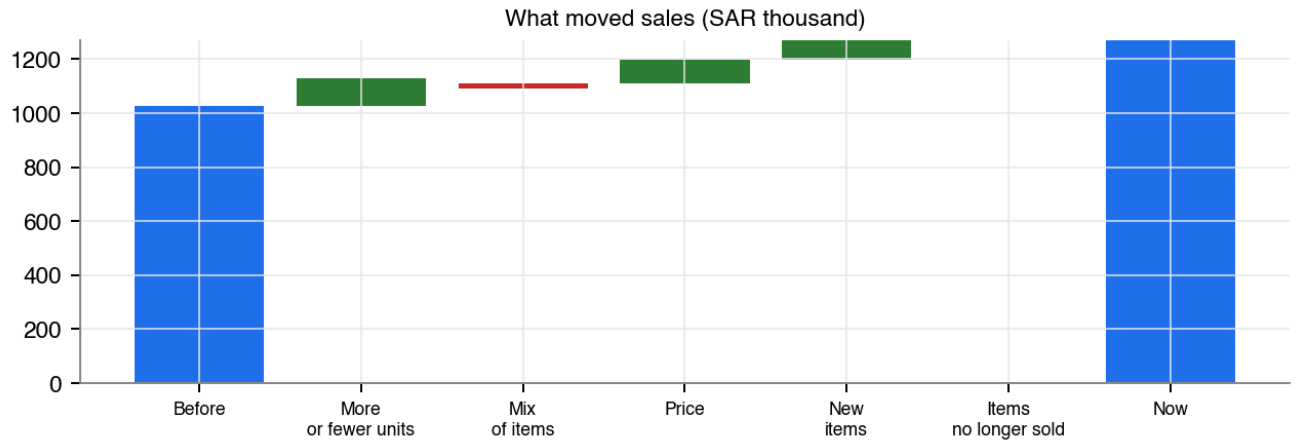


Figure 2. The change in sales, 2026-07 to 2026-09 against the same months a year earlier. Source: tables/drivers.csv

Step	SAR
Before	1,025,615
More or fewer units	102,272
Mix of items	-17,547
Price	85,688
New items	72,283
Items no longer sold	0
Now	1,268,311

Source: tables/drivers.csv

The parts add up to the change exactly: more or fewer units at the old prices, a shift towards dearer or cheaper dishes, price, and dishes that came or went. The dishes that moved most:

Dish	Before	Now	Change	Price
Water	9,976	11,797	1,821	-0.1%
Garden Salad	39,240	42,496	3,256	-0.0%
Soft Drink	16,986	20,388	3,402	+0.0%
Hummus	24,795	28,508	3,713	+0.0%
Grilled Chicken	132,251	156,666	24,415	+15.9%
Beef Burger	129,091	156,402	27,310	+14.7%
Family Platter	165,699	199,538	33,839	+0.1%
M06 (not in the item list)	0	72,283	72,283	n/a

Source: tables/drivers_items.csv

3. Profit by branch

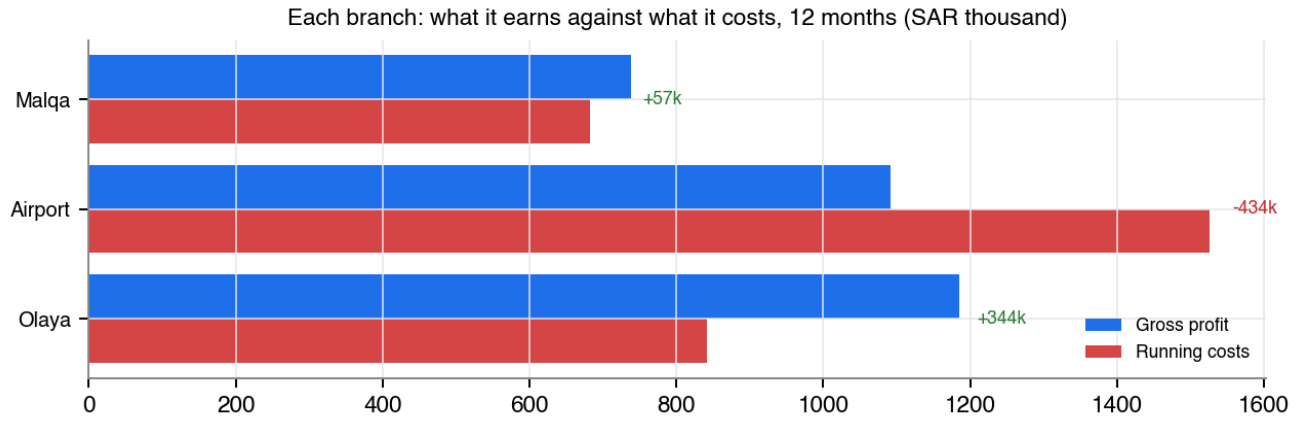


Figure 3. Gross profit against running costs, last 12 months. Source: tables/place_pnl.csv

Branch	Sales	Cost of sales	Gross profit	Delivery commissions	Head office	Maintenance	Marketing	Other operating	Rent	Salaries	Utilities	Opera profit
Olaya	1,966,331	780,612	1,185,719	90,238	0	25,188	0	36,159	240,000	375,941	73,736	344
Airport	1,795,188	703,457	1,091,731	84,597	0	27,101	0	37,227	900,000	406,421	70,501	-434
Malqa	1,218,130	479,319	738,811	58,023	0	28,974	0	39,244	144,000	339,783	71,728	57

Source: tables/place_pnl.csv

Head-office costs of SAR 362k come on top and belong to no single branch.

Compared fairly, by size:

Branch	Orders	Sales per seat a month	Sales per m² a month	Sales per staff hour	Share of sales
Olaya	26,237	2,048	546	94	39%
Airport	24,266	2,992	831	81	36%
Malqa	16,519	1,692	461	66	24%

Source: tables/place_pnl.csv

4. Dishes: what sells and what earns

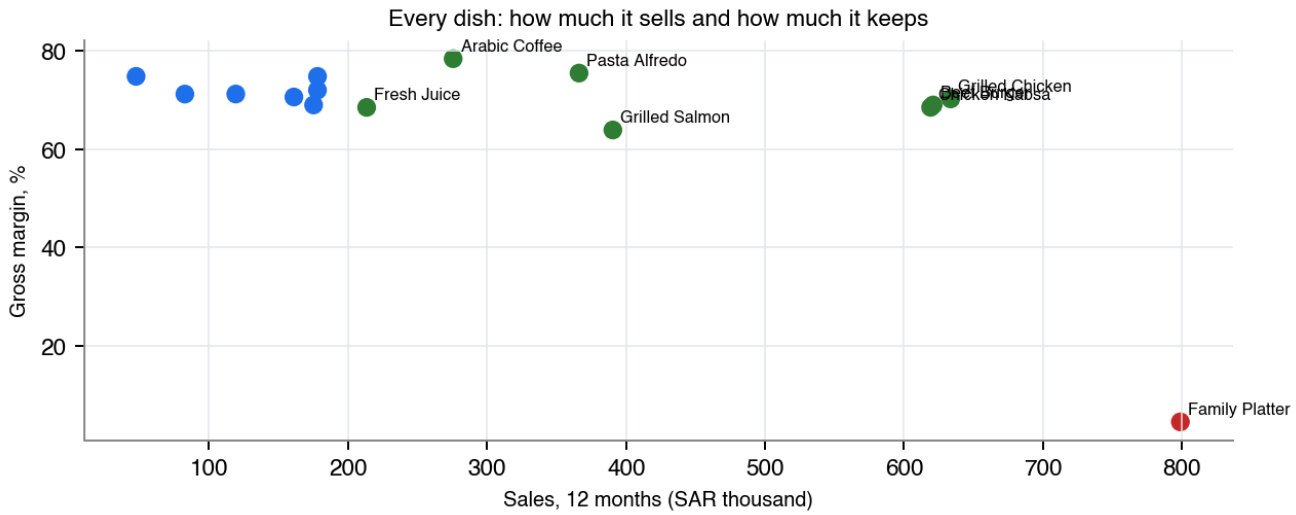


Figure 4. Sales against gross margin; red marks a popular dish with a low margin. Source: tables/items.csv

Dish	Category	Sales	Units	Margin	Share	Group
Family Platter	Sharing	798,808	8,476	4.5%	16.0%	Popular, low margin
Grilled Chicken	Mains	633,574	14,518	70.2%	12.7%	Strong
Beef Burger	Mains	620,896	16,053	69.0%	12.5%	Strong
Chicken Kabsa	Mains	619,123	13,003	68.5%	12.4%	Strong
Grilled Salmon	Mains	390,503	5,876	63.9%	7.8%	Strong
Pasta Alfredo	Mains	366,174	9,982	75.5%	7.4%	Strong
Arabic Coffee	Drinks	275,594	19,848	78.4%	5.5%	Strong
Fresh Juice	Drinks	213,374	13,449	68.5%	4.3%	Strong
Fries	Sides	178,021	14,960	74.8%	3.6%	Profitable, slow
Garden Salad	Sides	177,988	9,967	72.0%	3.6%	Profitable, slow
Kunafa	Desserts	175,196	6,793	69.0%	3.5%	Profitable, slow
Cheesecake	Desserts	161,009	6,763	70.6%	3.2%	Profitable, slow
M06 (not in the item list)	Not in the item list	120,120	3,669	n/a	2.4%	Cost unknown
Hummus	Sides	119,209	8,582	71.2%	2.4%	Profitable, slow
Soft Drink	Drinks	82,583	11,894	71.2%	1.7%	Profitable, slow
Water	Drinks	47,478	11,966	74.8%	1.0%	Profitable, slow

Source: tables/items.csv

5. Guests

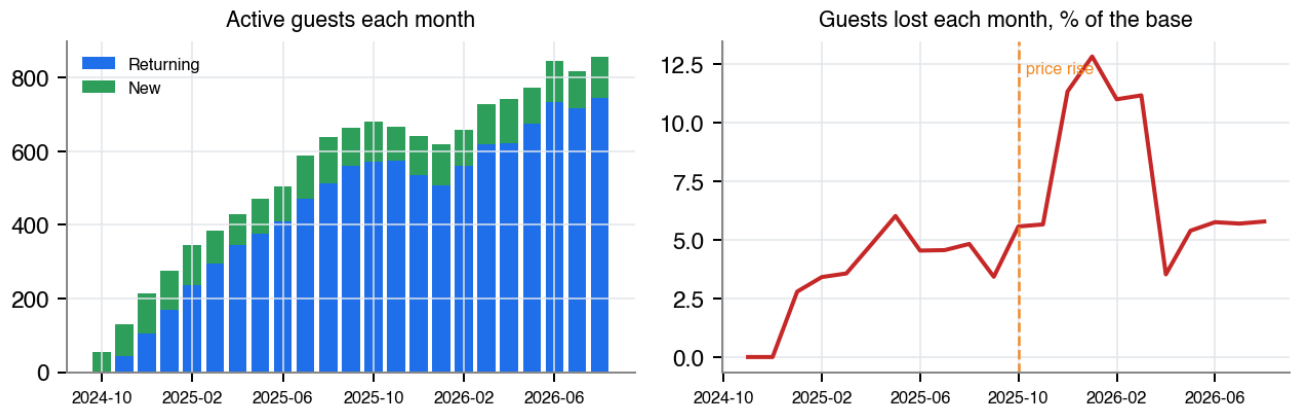


Figure 5. New and returning guests, and the share lost each month. Source: tables/customers_monthly.csv

Known guests account for 28% of sales; the rest pay without being identified. Of new ones, 89% come back within 90 days, and a new guest brings SAR 1,115 of sales in the first year.

Group	Guests	Sales	Average visits	Days since last (median)	Share of sales
At risk	36	86,878	32.8	68	4%
Lost	1219	909,901	10.4	321	37%
Loyal	372	1,014,448	37.0	6	41%
New	230	51,134	3.0	8	2%
Occasional	610	393,720	8.7	21	16%

Source: tables/segments.csv

6. How guests reacted to prices

Category	Month	Price	Units	Elasticity	Lost a month, before	Lost a month, after
Mains	2025-10	+15.9%	-6.9%	-0.43	4.3%	8.8%

Source: tables/price_events.csv

Elasticity is the change in units for each 1% of price: -0.5 means a 10% rise loses 5% of units. It is measured over the three months either side, against all orders, so it mixes the price with whatever else changed at the time.

7. Marketing

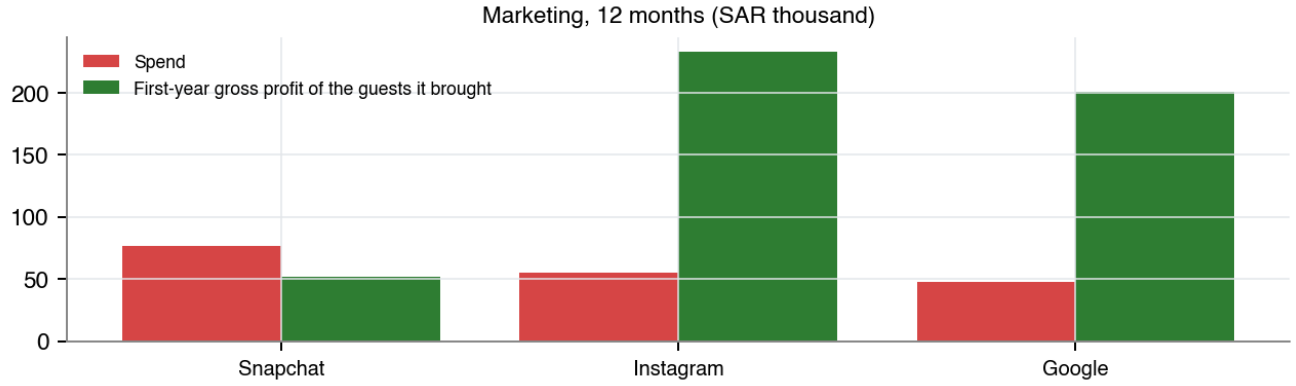


Figure 6. Spend against the first-year gross profit of the guests each channel brought. Source: tables/marketing.csv

Channel	Spend	Clicks	New guests	Cost of each	First-year gross profit	Return per unit
Snapchat	76,564	125,821	80	957	51,711	0.68
Instagram	54,731	89,731	360	152	232,699	4.25
Google	47,621	76,382	310	154	200,379	4.21

Source: tables/marketing.csv

A return below one means the channel cost more than its guests earned back in their first year. Guests who joined through no channel are not counted, so the returns are a floor.

8. People, hours and busy times

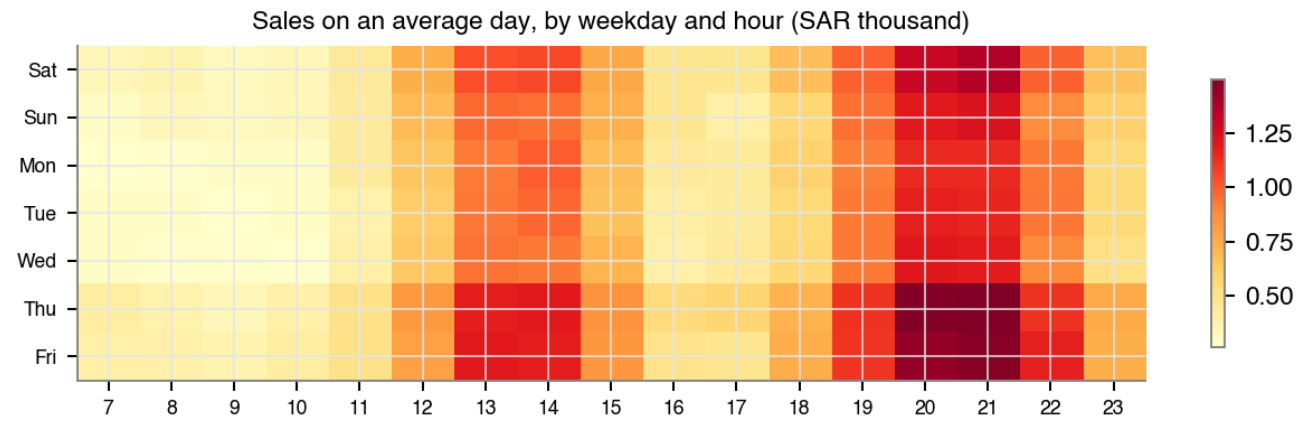


Figure 7. When sales happen. Source: tables/busy.csv

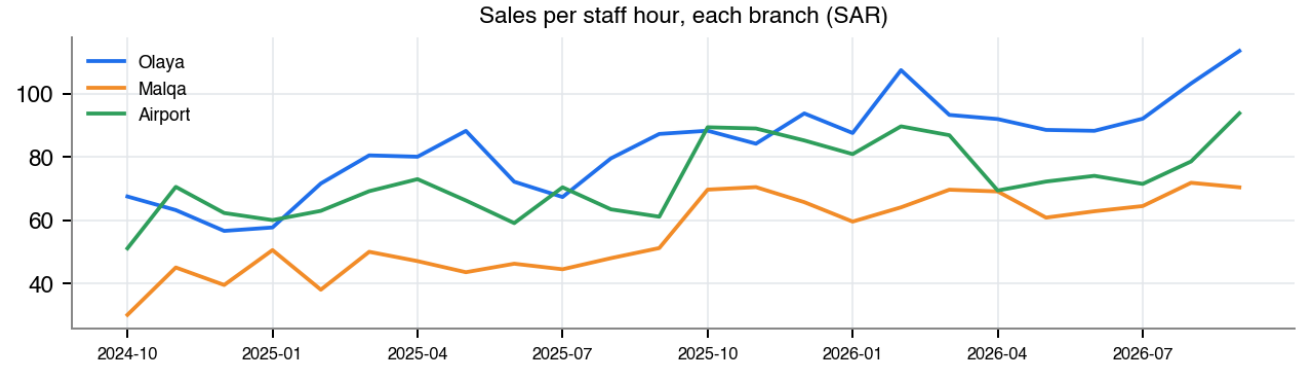


Figure 8. Sales per staff hour by branch. Source: tables/place_month.csv

The measures restaurants and cafés usually watch:

Measure	Airport	All	Malqa	Olaya
Average order value	74	74	74	75
Food cost	39.2%	39.4%	39.3%	39.7%
Orders per seat a day	1.33	0.97	0.75	0.90
Sales per labour hour	81	81	66	94
Sales per square metre a month	831	593	461	546

Source: tables/kpis.csv

9. Against target

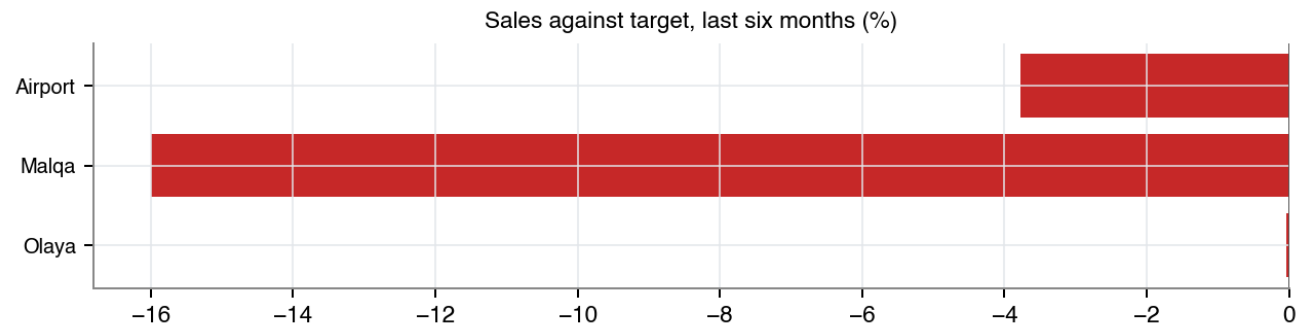


Figure 9. Sales against target over the last six months. Source: tables/targets.csv

Month	Branch	Measure	Target	Actual	Gap
2026-07	Olaya	revenue	165,900	166,977	+0.6%
2026-07	Malqa	revenue	123,200	102,448	-16.8%
2026-07	Airport	revenue	151,900	144,482	-4.9%
2026-08	Olaya	revenue	181,100	177,859	-1.8%
2026-08	Malqa	revenue	122,000	103,703	-15.0%
2026-08	Airport	revenue	152,900	146,105	-4.4%
2026-09	Olaya	revenue	171,900	173,434	+0.9%
2026-09	Malqa	revenue	120,700	102,264	-15.3%
2026-09	Airport	revenue	155,500	151,038	-2.9%
2026-07	Olaya	orders	2,134	2,200	+3.1%
2026-07	Malqa	orders	1,579	1,362	-13.7%
2026-07	Airport	orders	2,015	1,976	-1.9%
2026-08	Olaya	orders	2,289	2,360	+3.1%
2026-08	Malqa	orders	1,620	1,397	-13.8%
2026-08	Airport	orders	2,019	1,980	-1.9%
2026-09	Olaya	orders	2,220	2,289	+3.1%
2026-09	Malqa	orders	1,591	1,372	-13.8%
2026-09	Airport	orders	2,066	2,026	-1.9%

Source: tables/targets.csv

10. What a new branch would earn

Sized like the 3 existing branches: SAR 123k of sales a month once mature, a gross margin of 60.6%, rent of SAR 18k and staff of SAR 30k a month. That leaves SAR 7,755 a month when mature, paying back the fit-out in about 58 months. The existing branches range from SAR 1,692 to 2,992 of sales per seat a month, so location decides most of the answer. The dashboard lets the owner change every assumption.

11. The data received

Every line exported was read; what could not be used is listed here with the reason. The data readiness note says what to collect next, and how.

Source	Table	Field	What happened	Rows	Example
pos	transactions		rows read / kept	257,034	kept 251732
pos	transactions	status	excluded by rule: ['void']	5,302	
menu	items		rows read / kept	15	kept 15
branches	places		rows read / kept	3	kept 3
ledger	expenses		rows read / kept	480	kept 480
payroll	staff		rows read / kept	72	kept 72
ads	marketing		rows read / kept	2,190	kept 2190
loyalty	parties		rows read / kept	2,730	kept 2730
budget	targets		rows read / kept	144	kept 144
	transactions	item_id	not found in items	2,906	M06
	expenses	place_id	not found in places	48	HQ

Source: tables/intake_log.csv

Days on which a branch that trades every day has no data:

place_id	date	issue
B1	2026-02-14	no transactions on a day the site normally trades

Source: tables/data_gaps.csv

12. All health tests

Area	Test	Result	Threshold	Status
Data	Share of exported lines that could be used	97.9%	pass >= 97.0%; fail < 90.0%	PASS
Data	Days since the latest transaction	5	pass <= 10; fail > 35	PASS
Data	Trading days with no data	1	pass <= 0; fail > 3	WARN
Data	Sales whose cost is unknown	1.4%	pass <= 1.0%; fail > 5.0%	WARN
Profit	Operating margin after head office, last 12 months	-7.9%	pass >= 5.0%; fail < 0.0%	FAIL
Profit	Share of sales made at loss-making sites	36.1%	pass <= 0.0%; fail > 25.0%	FAIL
Profit	Gross margin, last 12 months	60.6%	pass >= 60.0%; fail < 50.0%	PASS
Items	Share of sales from items with a low margin	16.0%	pass <= 5.0%; fail > 15.0%	FAIL
Sales	Sales growth, recent months	23.7% (the same months a year earlier)	pass >= 0.0%; fail < -5.0%	PASS
Customers	New customers who come back within 90 days	89.1%	pass >= 40.0%; fail < 25.0%	PASS
Customers	Customers lost each month, change after the latest price rise	107.2% (Mains, 2025-10)	pass <= 20.0%; fail > 50.0%	FAIL
Customers	Units lost for each 1% of price rise	0.43 (Mains, 2025-10)	pass <= 0.50; fail > 1.00	PASS
Marketing	Worst channel: first-year gross profit per unit of spend	0.68 (Snapchat)	pass >= 1.00; fail < 0.50	WARN
Operations	Weakest site: sales per staff hour against the best	70.7%	pass >= 80.0%; fail < 60.0%	WARN
Planning	Site-months more than 5% below target, last 6 months	33.3%	pass <= 20.0%; fail > 50.0%	WARN
Planning	Forecast error on months held back	10.9%	pass <= 10.0%; fail > 20.0%	WARN
Sales	Share of sales from the largest site	39.5%	pass <= 50.0%; fail > 70.0%	PASS

Source: tables/tests.csv

Limitations and assumptions

- The analysis shows what happened and what moved together. It does not prove cause: a price change and a fall in guests may share another reason. A controlled test proves cause.
- Profit by branch is only as good as the way costs are booked to branches in the accounts.
- Guests who pay without being identified are outside the guest analysis.
- The forecast extends the past. It cannot see a new competitor, a lost contract or a change in the market.

Appendix

Preparation log

step	action	rows_affected
transactions	identical export lines removed	8121
transactions	after the period end 2026-09-30 left out	0
transactions	refunds kept as negative sales	512
transactions	cost taken from the item list (quantity x unit cost)	240774
transactions	no cost known (item missing from the item list)	2837
transactions	trading days with no data (listed)	1

Source: tables/cleaning_log.csv

Data quality

expenses · 480 rows · PASSED (0 errors, 0 warnings)

No data-quality issues detected.

items · 15 rows · PASSED (0 errors, 0 warnings)

No data-quality issues detected.

marketing · 2190 rows · PASSED (0 errors, 0 warnings)

No data-quality issues detected.

parties · 2730 rows · PASSED (0 errors, 1 warnings)

Check	Severity	Column	Count	Message
missing_values	warning	segment	2730	2730 missing (100.00%)

Source: data_quality/dq_parties.json

places · 3 rows · PASSED (0 errors, 0 warnings)

No data-quality issues detected.

staff · 72 rows · PASSED (0 errors, 0 warnings)

No data-quality issues detected.

targets · 144 rows · PASSED (0 errors, 0 warnings)

No data-quality issues detected.

transactions · 251732 rows · PASSED (0 errors, 8 warnings)

Check	Severity	Column	Count	Message
missing_values	warning	line_no	251732	251732 missing (100.00%)
missing_values	warning	party_id	181522	181522 missing (72.11%)
missing_values	warning	unit_price	251732	251732 missing (100.00%)
missing_values	warning	discount	251732	251732 missing (100.00%)
missing_values	warning	cost	251732	251732 missing (100.00%)
missing_values	warning	tax	251732	251732 missing (100.00%)
missing_values	warning	staff_id	251732	251732 missing (100.00%)
duplicate_rows	warning		8121	8121 fully duplicated rows

Source: data_quality/dq_transactions.json

transactions_ready · 243611 rows · PASSED (0 errors, 1 warnings)

Check	Severity	Column	Count	Message
missing_values	warning	cost	2837	2837 missing (1.16%)

Source: data_quality/dq_transactions_ready.json

Audit trail

Run ID	`ba_20261111T180200Z_d5dc74`
Run started / finished (UTC)	2026-11-11T18:02:00+00:00
Analyst	info@proquantanalytics.com
Code version (git)	`2703b230a84f (uncommitted changes) · core 0.1.0`
Config SHA-256	`7143263c7fcb21d5`
Input: pos	`257034 rows · sha256 7d843af5e446cb27`
Input: loyalty	`2730 rows · sha256 5cdd91850ba23aa6`
Input: budget	`72 rows · sha256 92fb307eeb3613a0`
Input: payroll	`72 rows · sha256 9ce7401c654058f1`
Input: menu	`15 rows · sha256 d9013d82a0d97fe6`
Input: ads	`2190 rows · sha256 2141427621ddf485`
Input: branches	`3 rows · sha256 7213bda7ecaa7296`
Input: ledger	`480 rows · sha256 5366f65dc6e5365a`
Environment	`python 3.13.5 · pandas 2.2.3 · numpy 2.1.3`

Source: audit.json in the run folder

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